

TuniLoon

High-Altitude Balloon Ground Station Software

Project Summary

August 10, 2026

Project Overview

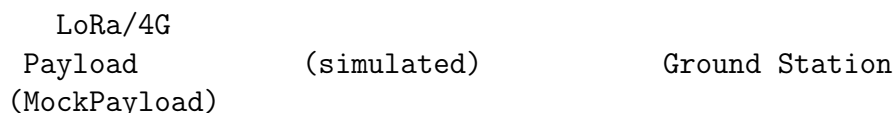
TuniLoon is a complete ground station software suite designed for high-altitude balloon missions. It simulates a balloon payload, receives telemetry, visualises data in real time, detects anomalies with machine learning, analyses historical flights, and supports multiple balloons simultaneously – all **without any hardware**.

Goal: Provide a robust, scalable platform for monitoring, analysing, and planning balloon flights, ready for future hardware integration.

Key Features

Feature	Description
Mock Payload	Realistic flight path (0→30000m→0), sensor simulation (BME680, MLX90640), and telemetry packet generation.
Live Dashboard	Interactive map (Leaflet), altitude/temperature graphs (Plotly), real-time telemetry, dark/light theme.
ML Anomaly Detection	Isolation Forest model trained on flight data; flags anomalies with confidence scores and Telegram alerts.
Flight Analysis	Compute metrics (duration, max altitude, ascent/descent rates, distance), estimate wind from GPS drift, generate Markdown reports.
Multi-Balloon	Track multiple balloons on one map; each with its own colour, path, and details sidebar.
Weather Integration	Fetch current weather (OpenWeatherMap) and display it on the map.
Cloud Connectivity	Publish telemetry to MQTT (HiveMQ), receive Telegram alerts (landing, peak altitude, anomalies), upload to Sondehub (public tracking).
Full Test Suite	Unit, integration, and stress tests covering decoders, loggers, ML models, and analysis.

Architecture



- Flight Path
 - Sensors
 - Telemetry
 - Anomalies
- Dashboard
 - Map
 - Graphs
 - Wind est.
- Analysis
 - Metrics
 - Reports
- Cloud Integration
 - MQTT, Telegram, Sondehub
- ML Module
 - Feature extraction
 - Isolation Forest
 - Real-time inference

Technologies

Layer	Stack
Backend	Python 3.10+, Flask, scikit-learn, joblib
Frontend	JavaScript, Leaflet.js, Plotly, HTML5/CSS3
Data	CSV, JSON, SQLite (optional)
Cloud	MQTT (HiveMQ), Telegram Bot API, Sondehub API
Testing	unittest, pytest

Sprints Completed

Sprint	Focus	Status
Sprint 0	Foundation (protocol, config)	Skipped
Sprint 1	Mock Payload Generator	
Sprint 2	Ground Station Core (decoder, logger)	
Sprint 3	Web Dashboard	
Sprint 4	Cloud Integration (MQTT, Telegram, Sondehub)	
Sprint 5	Testing & Simulation	
Sprint 6	Mobile App	
Sprint 7	ML Anomaly Detection	
Sprint 8	Historical Data Analysis	
Sprint 9	Multi-Balloon Support	
Sprint 10	Weather Integration	

Setup & Installation

1. Clone the repository

```
git clone https://github.com/TuniLoon/Software.git
cd Software
```

2. Create conda environment

```
conda create -n tuniloon python=3.10 -y
conda activate tuniloon
```

3. Install dependencies

```
pip install -r requirements.txt
```

4. (Optional) Configure Telegram & OpenWeatherMap

- Edit config/cloud_config.json with your Telegram bot token and chat ID.
- Edit config/ml_config.json to adjust anomaly thresholds.
- Add your OpenWeatherMap API key in ground_station/src/weather_service.py.

5. Run the dashboard

```
python -m ground_station.web.app
```

Open <http://localhost:5000> – the main dashboard will start automatically with mock data.

Usage

- **Main Dashboard:** / – live tracking, graphs, anomalies.
- **Analysis:** /analysis – flight metrics, wind estimation, report download.
- **Multi-Balloon:** /multi – track multiple balloons (add demo balloons with the "+ Demo" button).

Testing

```
# Run all unit tests
python -m unittest discover tests

# Run a specific test
python tests/test_decoder.py
```

Future Work (Without Hardware)

- **Flight Planner:** Simulate launch conditions and landing predictions.
- **Historical Data Browser:** Filter and compare past flights.
- **Automated PDF Reports:** Generate detailed reports automatically.
- **Real-time Simulation:** Run the simulator in wall-clock time.
- **KML/GPX Export:** Download flight paths for external tools.

Team

- **Saif Maammar** – Product Owner
- **Amal Karaoud** – Scrum Master
- **Mahdi BenSlima** – Payload Lead
- **Nesrine Zouari** – Ground Station Lead

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Made with  by the TuniLoon Team